

Sofia Marchetti

Senior Software Engineer | ML Systems & Platform Infrastructure

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CORE SKILLS

Languages & Frameworks: Python, Go, Java, TypeScript | **ML/LLM:** LangChain, Claude API, vector databases, prompt engineering | **Cloud & Infrastructure:** AWS (EC2, RDS, Lambda, SageMaker), Kubernetes, Docker, Terraform | **Data:** PostgreSQL, Elasticsearch, Apache Kafka, DuckDB | **Tools & Practices:** CI/CD pipelines, monitoring (Datadog), API design, microservices architecture

PROFESSIONAL EXPERIENCE

Senior Software Engineer, State of New York - Department of Labor

Albany, NY | March 2024 – Present

- Architected and deployed an LLM-powered claims processing agent serving 40,000+ monthly users; integrates Claude API with legacy COBOL systems via REST layers and handles asynchronous job workflows
- Reduced manual claim intake processing by 65% through rule-based agent design; system now handles unemployment insurance questions, form prefilling, and document classification with human review gates
- Engineered real-time document embedding pipeline (5M+ historical PDFs) using vector database for policy retrieval; maintains 94% uptime across three AWS regions
- Collaborated with policy teams and field offices to establish agent response quality benchmarks; implemented monitoring dashboards tracking accuracy, latency, and user satisfaction metrics
- Led API standardization effort connecting legacy government databases to modern inference endpoints, enabling secure credential handling for PII-sensitive workflows

Staff Backend Engineer, MetroTransit Authority

Buffalo, NY | June 2021 – February 2024

- Owned core platform services for real-time transit data serving 2M+ residents; designed and operated PostgreSQL clusters, Kafka topics, and microservices handling 15K req/sec peak load
- Built data platform ingesting GPS, ridership, and maintenance logs from 3,000+ vehicles; established Elasticsearch pipelines enabling city planners and operations teams to identify service gaps
- Implemented automated alert system (Go, Kubernetes) that detected service degradation 10 minutes before passenger impact; reduced mean time to recovery by 40%
- Mentored team of five engineers on production Kubernetes operations and disaster recovery procedures; established runbooks adopted across infrastructure group

Backend Engineer, Government Technology Solutions Inc.

Rochester, NY | September 2018 – May 2021

- Shipped public records search API (PostgreSQL, Node.js, AWS Lambda) for three state agencies; maintained 99.9% uptime serving 500K queries/month
- Designed and deployed containerized ETL pipelines processing daily document uploads; reduced data ingestion latency from 8 hours to 45 minutes through Kafka-based architecture
- Implemented OAuth 2.0 integration layer allowing secure third-party access to restricted citizen datasets; security review certified for HIPAA and FERPA compliance
- Created monitoring infrastructure (CloudWatch, custom metrics) and on-call rotation documentation; trained operations staff on deployment procedures

Software Engineer, Regional Data Initiatives

Troy, NY | July 2017 – August 2018

- Built internal tools for census data aggregation and visualization; maintained Java backend and PostgreSQL warehouse supporting 200+ government analysts
- Developed API documentation and conducted training sessions for downstream teams consuming data services

EDUCATION

B.S. Computer Science, State University of New York at Buffalo | 2017